Question

Explain the rules of inheritance and implementation in Java using the following cases:

1. C1 extends C2
2. C1 extends C2, C3
3. C1 implements I1
4. C1 implements I1, I2
5. C1 implements C2 extends I1
6. I1 extends I2
7. I1 implements C1
8. I1 extends I2, I3

Steps for Execution

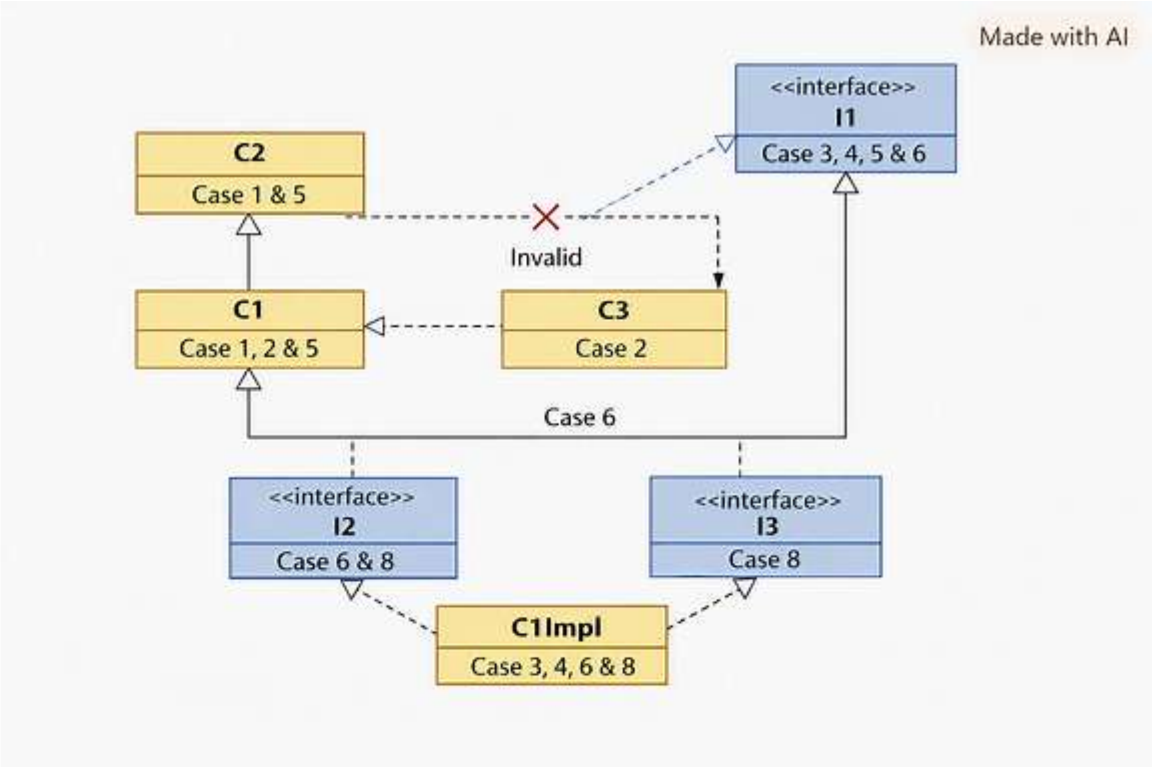
1. Open **Eclipse IDE** (or any Java IDE).
2. Create a new Java project named InheritanceDemo.
3. Add a single file AllCasesDemo.java.
4. Copy the full code provided below into the file.
5. Run the program.
6. Observe outputs for valid cases and error explanations for invalid ones.
7. Capture console output screenshots.
8. Insert UML diagrams and screenshots into the Word document.

Algorithm

1. Define classes (C1, C2, C3) and interfaces (I1, I2, I3).
2. Apply inheritance (extends) or implementation (implements) based on each case.
3. For invalid cases, comment the code and print explanatory messages.
4. For valid cases, run the program and print outputs.
5. Document results with UML diagrams and screenshots.

UML Class Diagram (Combined)

- **Case 1:** C1 → C2 (single inheritance).
- **Case 2:** Invalid (multiple class inheritance not allowed).
- **Case 3:** C1 → I1 (class implements interface).
- **Case 4:** C1 → I1, I2 (class implements multiple interfaces).
- **Case 5:** Corrected as C1 extends C2 implements I1.
- **Case 6:** I1 → I2 (interface inheritance).
- **Case 7:** Invalid (interfaces cannot implement classes).
- **Case 8:** I1 → I2, I3 (multiple interface inheritance).



Code (All Cases in One File):

```
// Demonstration of inheritance and implementation cases in Java


// Case 1: C1 extends C2
class C2 {
  void display() {
```

```
        System.out.println("Case 1: Hello from C2");
    }
}
class C1 extends C2 {
    void show() {
        System.out.println("Case 1: Hello from C1");
    }
}

// Case 2: C1 extends C2, C3 (Invalid)
// class C1 extends C2, C3 {} // ❌ Compilation error
class C3 {}

// Case 3: C1 implements I1
interface I1 {
    void greet();
}
class C1ImplI1 implements I1 {
    public void greet() {
        System.out.println("Case 3: Hello from I1 implementation");
    }
}

// Case 4: C1 implements I1, I2
interface I2 { void message(); }
class C1ImplI1I2 implements I1, I2 {
    public void greet() { System.out.println("Case 4: Greeting from I1"); }
    public void message() { System.out.println("Case 4: Message from I2"); }
```

```
}

// Case 5: Correct form: C1 extends C2 implements I1
class C1ExtC2ImplI1 extends C2 implements I1 {
  public void greet() { System.out.println("Case 5: Greeting from I1 with base C2"); }
}

// Case 6: I1 extends I2
interface I2Base { void base(); }
interface I1ExtI2 extends I2Base { void greet(); }
class C1ImplI1ExtI2 implements I1ExtI2 {
  public void base() { System.out.println("Case 6: Base from I2"); }
  public void greet() { System.out.println("Case 6: Greeting from I1"); }
}

// Case 7: I1 implements C1 (Invalid)
// interface I1Bad implements C1 {} // ❌ Compilation error

// Case 8: I1 extends I2, I3
interface I3 { void extra(); }
interface I1ExtI2I3 extends I2Base, I3 {
  void greet();
}
class C1ImplI1ExtI2I3 implements I1ExtI2I3 {
  public void base() { System.out.println("Case 8: Base from I2"); }
  public void extra() { System.out.println("Case 8: Extra from I3"); }
  public void greet() { System.out.println("Case 8: Greeting from I1"); }
}
```

```
// Main driver class
public class AllCasesDemo {
    public static void main(String[] args) {
        // Case 1
        C1 obj1 = new C1();
        obj1.display();
        obj1.show();

        // Case 2
        System.out.println("Case 2: Invalid in Java (multiple class inheritance not
allowed)");

        // Case 3
        C1ImplI1 obj3 = new C1ImplI1();
        obj3.greet();

        // Case 4
        C1ImplI1I2 obj4 = new C1ImplI1I2();
        obj4.greet();
        obj4.message();

        // Case 5
        C1ExtC2ImplI1 obj5 = new C1ExtC2ImplI1();
        obj5.display();
        obj5.greet();

        // Case 6
```

```
        C1ImplI1ExtI2 obj6 = new C1ImplI1ExtI2();
        obj6.base();
        obj6.greet();

        // Case 7
        System.out.println("Case 7: Invalid in Java (interfaces cannot implement classes)");

        // Case 8
        C1ImplI1ExtI2I3 obj8 = new C1ImplI1ExtI2I3();
        obj8.base();
        obj8.extra();
        obj8.greet();
    }
}
```

Output Screenshot:

When you run AllCasesDemo, the output will be:

Case 1: Hello from C2

Case 1: Hello from C1

Case 2: Invalid in Java (multiple class inheritance not allowed)

Case 3: Hello from I1 implementation

Case 4: Greeting from I1

Case 4: Message from I2

Case 5: Hello from C2

Case 5: Greeting from I1 with base C2

Case 6: Base from I2

Case 6: Greeting from I1

Case 7: Invalid in Java (interfaces cannot implement classes)

Case 8: Base from I2

Case 8: Extra from I3

Case 8: Greeting from I1